

## KYUSHU SAGA INTL, Japan

WMO: 478100

Lat: 33.150N

Lon: 130.302E

Elev: 16

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>28.2 | (c)<br>30.2 | (d)<br>19.1                   | (e)<br>14.4 | (f)<br>36.7 | (g)<br>21.3 | (h)<br>16.0 | (i)<br>36.6 | (j)<br>23.1           | (k)<br>42.0 | (l)<br>21.0 | (m)<br>42.5 | (n)<br>5.1            | (o)<br>10 | (p)<br>0.444 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>12.5            | (c)<br>92.9     | (d)<br>79.9 | (e)<br>91.0 | (f)<br>79.4 | (g)<br>88.2 | (h)<br>78.8 | (i)<br>81.6         | (j)<br>88.1 | (k)<br>80.6 | (l)<br>87.1 | (m)<br>79.9 | (n)<br>86.0 | (o)<br>8.8           | (p)<br>230 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 79.2                            | 151.8 | 85.6 | 78.9 | 150.2 | 85.2 | 78.4 | 147.5 | 84.7 | 45.4          | 89.1 | 44.4 | 88.1 | 43.6 | 86.0 | 85.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |          | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|----------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |          | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max      | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 21.9          | (b) 19.1 | (c) 17.1 | (e) 23.1                   | (f) 95.3 | (g) 1.8            | (h) 0.7  | (i) 21.8                                           | (j) 95.7  | (k) 20.8   | (l) 96.1  | (m) 19.8   | (n) 96.5  | (o) 18.5   | (p) 97.0  |    |    |
| (4) 21.9          | (5) 19.1 | (6) 17.1 | (7) 23.1                   | (8) 95.3 | (9) 1.8            | (10) 0.7 | (11) 21.8                                          | (12) 95.7 | (13) 20.8  | (14) 96.1 | (15) 19.8  | (16) 96.5 | (17) 18.5  | (18) 97.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 61.7   | 41.1  | 43.8  | 49.5  | 57.9  | 66.9  | 73.8  | 81.3  | 83.2  | 76.2  | 66.0  | 55.0  | 44.9  |  |  |
|                                                                        | DBStd       | 15.14  | 4.43  | 5.63  | 5.59  | 5.06  | 4.03  | 3.47  | 3.63  | 3.14  | 4.36  | 4.85  | 5.97  | 5.51  |  |  |
|                                                                        | HDD50       | 745    | 276   | 189   | 79    | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 20    | 177   |  |  |
|                                                                        | HDD65       | 3032   | 740   | 595   | 482   | 218   | 26    | 0     | 0     | 0     | 0     | 45    | 303   | 623   |  |  |
|                                                                        | CDD50       | 5025   | 2     | 14    | 62    | 240   | 524   | 714   | 970   | 1028  | 785   | 497   | 170   | 19    |  |  |
|                                                                        | CDD65       | 1836   | 0     | 0     | 0     | 4     | 85    | 264   | 505   | 563   | 335   | 77    | 3     | 0     |  |  |
|                                                                        | CDH74       | 16221  | 0     | 0     | 0     | 9     | 372   | 1346  | 5170  | 6414  | 2537  | 371   | 2     | 0     |  |  |
|                                                                        | CDH80       | 5356   | 0     | 0     | 0     | 0     | 39    | 195   | 1764  | 2622  | 710   | 26    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 7.7    | 7.0   | 7.9   | 8.2   | 8.0   | 7.9   | 7.8   | 8.7   | 7.8   | 7.3   | 7.5   | 6.7   | 7.1   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 76.1   | 2.5   | 3.2   | 4.9   | 6.1   | 7.1   | 13.2  | 13.0  | 8.9   | 6.9   | 3.6   | 3.9   | 2.6   |  |  |
|                                                                        | PrecMax     | 104.1  | 7.0   | 6.2   | 11.0  | 12.2  | 14.4  | 21.5  | 24.8  | 23.6  | 21.5  | 7.8   | 7.8   | 5.6   |  |  |
|                                                                        | PrecMin     | 40.2   | 1.0   | 1.2   | 2.5   | 2.3   | 2.1   | 1.9   | 1.1   | 2.0   | 2.8   | 0.4   | 0.6   | 0.2   |  |  |
|                                                                        | PrecStd     | 14.2   | 1.3   | 1.4   | 1.8   | 2.6   | 3.1   | 5.7   | 6.7   | 5.2   | 4.1   | 2.3   | 2.0   | 1.4   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 57.4  | 63.0  | 68.3  | 75.4  | 84.1  | 86.4  | 93.6  | 95.3  | 91.4  | 82.6  | 72.0  | 63.0  |  |  |
|                                                                        |             | MCWB   | 53.1  | 56.4  | 59.8  | 61.8  | 67.5  | 74.5  | 80.1  | 80.1  | 78.2  | 71.7  | 64.0  | 56.2  |  |  |
|                                                                        | 2%          | DB     | 54.0  | 59.3  | 64.7  | 71.9  | 80.5  | 84.2  | 91.5  | 93.4  | 88.2  | 80.4  | 70.1  | 59.4  |  |  |
|                                                                        |             | MCWB   | 48.9  | 53.9  | 56.8  | 61.2  | 66.9  | 73.5  | 79.6  | 80.0  | 76.7  | 70.6  | 62.0  | 54.0  |  |  |
|                                                                        | 5%          | DB     | 52.0  | 57.1  | 62.4  | 69.8  | 78.4  | 82.3  | 89.7  | 91.6  | 86.2  | 78.5  | 68.1  | 57.2  |  |  |
|                                                                        |             | MCWB   | 46.5  | 52.0  | 56.4  | 60.8  | 66.1  | 72.9  | 79.2  | 79.8  | 75.8  | 68.6  | 61.4  | 51.7  |  |  |
|                                                                        | 10%         | DB     | 50.0  | 53.7  | 59.4  | 67.8  | 75.4  | 80.5  | 87.7  | 89.8  | 84.2  | 75.5  | 66.0  | 55.1  |  |  |
|                                                                        |             | MCWB   | 44.9  | 48.5  | 53.8  | 60.0  | 65.6  | 73.2  | 79.3  | 79.2  | 74.8  | 66.4  | 59.9  | 50.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 54.8  | 59.5  | 62.2  | 67.2  | 72.7  | 79.3  | 82.7  | 83.0  | 80.2  | 76.5  | 67.5  | 59.4  |  |  |
|                                                                        |             | MCDB   | 56.5  | 61.4  | 65.9  | 71.0  | 77.0  | 83.0  | 88.9  | 90.2  | 86.9  | 80.3  | 70.4  | 60.9  |  |  |
|                                                                        | 2%          | WB     | 51.4  | 55.7  | 59.3  | 65.0  | 70.6  | 77.9  | 81.6  | 81.8  | 79.1  | 73.9  | 65.1  | 55.4  |  |  |
|                                                                        |             | MCDB   | 53.7  | 58.8  | 63.2  | 69.1  | 75.3  | 81.1  | 87.4  | 88.9  | 85.2  | 78.1  | 68.4  | 58.5  |  |  |
|                                                                        | 5%          | WB     | 48.1  | 52.6  | 57.2  | 63.1  | 69.1  | 76.4  | 80.5  | 81.0  | 77.9  | 70.9  | 62.9  | 53.0  |  |  |
|                                                                        |             | MCDB   | 50.8  | 56.0  | 61.3  | 67.4  | 74.2  | 79.5  | 86.3  | 87.9  | 83.8  | 75.6  | 66.8  | 56.5  |  |  |
|                                                                        | 10%         | WB     | 45.5  | 49.8  | 54.9  | 61.3  | 67.6  | 74.7  | 79.8  | 80.1  | 76.7  | 68.1  | 60.3  | 50.7  |  |  |
|                                                                        |             | MCDB   | 49.0  | 52.8  | 59.2  | 65.8  | 73.3  | 78.1  | 85.3  | 86.8  | 82.3  | 73.0  | 64.7  | 54.0  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 15.1  | 15.2  | 16.1  | 17.2  | 17.0  | 11.6  | 10.1  | 12.5  | 14.2  | 17.0  | 16.7  | 15.4  |  |  |
|                                                                        |             | MCDBR  | 17.4  | 17.9  | 19.3  | 20.0  | 20.7  | 15.3  | 14.0  | 15.4  | 16.5  | 19.0  | 17.8  | 17.5  |  |  |
|                                                                        |             | MCWBR  | 12.9  | 13.5  | 13.5  | 12.5  | 11.2  | 7.8   | 6.6   | 6.6   | 7.7   | 10.0  | 11.4  | 13.0  |  |  |
|                                                                        | 5% WB       | MCDBR  | 14.8  | 15.7  | 15.2  | 14.9  | 15.3  | 8.5   | 11.2  | 12.7  | 12.7  | 12.5  | 13.6  | 14.7  |  |  |
|                                                                        |             | MCWBR  | 12.8  | 13.6  | 12.4  | 11.1  | 9.8   | 5.7   | 5.9   | 6.2   | 6.8   | 8.3   | 10.5  | 12.3  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.439 | 0.481 | 0.543 | 0.562 | 0.572 | 0.584 | 0.573 | 0.544 | 0.514 | 0.466 | 0.436 | 0.431 |  |  |
|                                                                        | taud        |        | 2.031 | 1.959 | 1.814 | 1.821 | 1.858 | 1.911 | 1.982 | 2.055 | 2.092 | 2.127 | 2.124 | 2.069 |  |  |
|                                                                        | Ebn at Noon |        | 235   | 239   | 234   | 237   | 236   | 233   | 234   | 239   | 240   | 241   | 235   | 229   |  |  |
|                                                                        | Edh at Noon |        | 44    | 52    | 65    | 67    | 65    | 62    | 57    | 52    | 49    | 43    | 40    | 40    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 703   | 929   | 1270  | 1525  | 1641  | 1380  | 1561  | 1637  | 1333  | 1125  | 831   | 652   |  |  |
|                                                                        | RadStd      |        | 70    | 89    | 84    | 132   | 187   | 191   | 222   | 163   | 139   | 116   | 81    | 47    |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (30 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page